Year 13 Maths - Sequences and series

Similar questions - Geometric progressions and convergence

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A geometric progression has first term 80 and fourth term 5.12.

(a) Find the common ratio and the sum to infinity. **[4]**

(b) Find the sum of the first 5 terms. **[2]**

(c) Find the least value of n for which the sum to infinity exceeds the sum of the first n terms by less than 0.01. **[4]**

2 The numbers $x+4$, $2x-4$ and $x-1$ are the first three terms of a geometric progression.

(a) Show that $3x^2 - 19x + 20 = 0$. **[3]**

(b) Find the two possible values of x and, for each, write down the first three terms of the progression. **[3]**

(c) Show that both progressions converge and find the sum to infinity of each. **[4]**

Question 1

(a) $80r^3 = 5.12$, so $r^3 = 0.064$ and $r = 0.4$. $S_\infty = \dfrac{80}{1-0.4} = 133.\overline{3} = \dfrac{400}{3}$

(b) $S_5 = \dfrac{80(1-0.4^5)}{0.6} = 131.968$

(c) $S_\infty - S_n = \dfrac{400}{3}(0.4)^n < 0.01$, so $0.4^n < 7.5 \times 10^{-5}$ and $n > \dfrac{\ln(7.5 \times 10^{-5})}{\ln 0.4} = 10.4$. The least n is 11

Question 2

(a) $\dfrac{2x-4}{x+4} = \dfrac{x-1}{2x-4}$, so $(2x-4)^2 = (x+4)(x-1)$, giving $4x^2 - 16x + 16 = x^2 + 3x - 4$ and $3x^2 - 19x + 20 = 0$

(b) $(3x-4)(x-5) = 0$: $x = \frac{4}{3}$ or $x = 5$. For $x = 5$: $9,\ 6,\ 4$. For $x = \frac{4}{3}$: $\frac{16}{3},\ -\frac{4}{3},\ \frac{1}{3}$

(c) $x = 5$: $r = \frac{2}{3}$, $S_\infty = \dfrac{9}{1/3} = 27$. $x = \frac{4}{3}$: $r = -\frac{1}{4}$, $S_\infty = \dfrac{16/3}{5/4} = \dfrac{64}{15}$. Both have $|r| < 1$, so both converge